THE UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

Kiểm thử phần mềm - 22KTPM3

AUTOMATION TESTING - HW5

Thành Viên

ID	Full name
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Chapter 1

Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	- SignUp	Done
		- Checkout	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
		- User Management	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

Chapter 2

Automation Testing

2.1 Automation process step by step

2.1.1 Test Case Creation

- A test case was recorded and created using the Web Recorder in Katalon Studio to test sorting behavior on a web page.
- Link: Catalog, Categories

2.1.2 Element Identification

- Web elements like dropdowns, buttons, and result containers were captured and stored in the Object Repository.

2.1.3 Parameterization

- A variable called sort_value, isRegex, search_value was declared and mapped to external test data for data-driven execution.

2.1.4 Test Data Binding

- A CSV/Excel file was created with different sorting options and different search value

2.1.5 Loop Execution

- The test case was executed once per row of data using a test suite to verify each sorting option.

2.2 Data-Driven Testing Implementation

2.2.1 For Catalog Test case:

Test Data File:

- File format: Excel
- Content example:

sort_value	isRegex	search_value
name,asc	TRUE	aaa
name,desc	TRUE	#$@%!
price,asc	TRUE	aaa
price,desc	TRUE	drill123

- Data Binding Steps:

1. **Create Test Data:** In Data Files, create new test data with file data we have (browse file).
2. **Create Variable in Test Case:** In test case, go to Variables tab, add variable sort_value (type: String), isRegex(type: boolean), search_value(type: String).
3. **Data Binding:** Open Data Binding, bind sort_value to sort_value column of test data, ...

2.2.2 For Categories Test case:

Test Data File:

- File format: Excel
- Content example:

sort_value	isRegex
name,asc	TRUE
name,desc	TRUE
price,asc	TRUE
price,desc	TRUE

- Data Binding Steps:

1. **Create Test Data:** In Data Files, create new test data with file data we have (browse file).

2. **Create Variable in Test Case:** In test case, go to Variables tab, add variable sort_value (type: String), isRegex(type: boolean).
3. **Data Binding:** Open Data Binding, bind sort_value to sort_value column of test data, ...

2.3 Use of Checkpoints / Assertions / Verifications

2.3.1 For Catalog Test case:

- In the Catalog Test Case, we use a simple element verification to ensure the expected UI element (catalog navigation link) is present after navigation and filtering interactions. This serves as a lightweight checkpoint to confirm page stability and visibility of a key element.

```
// Verification
WebUI.verifyElementPresent(findTestObject('Object Repository/Page_Practice Software Testing - Toolshop -_8413b3/a_Hand Tools_nav-link active'), 10, FailureHandling.STOP_ON_
```

Figure 2.1: Verifications for Catalog

- This checkpoint ensures that after clicking and navigating through menus, the correct page (Catalog/Hand Tools section) is still active.
- It adds a safety net without over-verifying UI or product data.
- Helps catch broken links or missing navigation tabs early in the process.

2.3.2 For Categories Test case:

- In the Categories Test Case, we include a verification step to ensure that a specific category checkbox (e.g., Screwdriver) is selected after the user interacts with the category filter on the interface.

```
// Verification
WebUI.verifyElementChecked(findTestObject('Object Repository/Page_Practice Software Testing - Toolshop -_8413b3/input_Screwdriver_category_id'), 10, FailureHandling.STOP_ON
```

Figure 2.2: Verifications for Categories

- This ensures that the category selection action was successfully registered on the UI.
- It validates that the category filter logic works correctly when the user selects an option.
- This type of verification is essential for catching UI bugs or user interaction issues that could impact the product browsing experience.

2.4 Test Results

- **Execution Platform:** Chrome, Edge , Safari
- **Data Sets**: Two file Excel
- **Test Status**:

Test Case	Browser	Status
Categories	Chrome	Passed
Categories	Edge	Passed
Categories	Safari	Fail
Catalog	Chrome	Passed
Catalog	Edge	Passed
Catalog	Safari	Fail

2.5 Demo video

2.5.1 Feature Catalog:

- Video demo Automation Testing for Catalog

2.5.2 Feature Categories:

- Video demo Automation Testing for Categories

Chapter 3

Self-Evaluation

Criteria	Max Point	Self-assessment
1. Catalog	50	50
1.1. Report + Bug	15	15
1.2 Test cases	5	5
1.3. Scripts files	10	10
1.4 Data files	10	10
1.5 Videos	10	10
2. Categories	50	50
2.1. Report + Bug	15	15
2.2 Test cases	5	5
2.3. Scripts files	10	10
2.4 Data files	10	10
2.5 Videos	10	10